How can we help?

Enter question or keyword. Example: Order Search Help

Back
My Orders
My Account
Balance & Coupons
Shipping & Delivery
Service & Refunds
Payments
Technical Support
Generate Gerber
Design Instruction
Ordering Parameter Instruction
Production Issues
Process Introduction
PCB Engineering Questions
PCB Assembly FAQ
Certification
3D Printing
CNC Machining
Sheet Metal
Injection Molding & Vacuum Casting
Design Service

Printed Circuit Board Terminology

As a new PCB designer, you may encounter many specialized terminologys related to printed circuit boards, whic often be confusing. Engineering courses typically focus more on system design and semiconductors rather than design, so they offer limited help in this area. However, the circuit board is the crucial link between these fields, m essential to master basic terminology.

To help new designers better understand printed circuit board terminology, we have compiled a list of common terminologys. By familiarizing yourself with these terminologys, you will be able to communicate more effectively colleagues and gain a deeper understanding of the design and manufacturing process of printed electronics. We introduce the terminology of printed circuit boards from the following three aspects:

Terminologys describing the physical structure of PCBs

Common terminologys used in PCB design

Terminologys related to manufacturing

PCB Physical Structure Terminologys

Some key printed circuit board terminologys are commonly used to describe the physical structure of PCBs, so understanding them is very important.

Layers: A circuit board is composed of multiple layers that are laminated together to form a single unit. Each laye etched copper to create the conductive paths on each layer's surface.

Copper Pour: In a PCB, there are some areas filled with copper, and the shapes of these areas can vary.

Traces and transmission lines: These two terms can be used interchangeably in advanced high-speed PCB desig referring to the conductive paths.

Signal vs. plane layer: The signal layer is mainly used for transmitting electrical signals, but it can also contain co areas for ground or power. The plane layer is used for distributing power or ground, and it is usually a complete co plane.

Vias: Vias are used to connect different layers of circuit boards, facilitating the transmission of electrical signals power connections, and aiding in the realization of complex circuit layouts and designs. They can be through-hole blind vias, or buried vias.

Components: Refers to any part placed on a PCB, including basic components like resistors, connectors, integrat circuits, and more. Components can be mounted by being soldered to the surface (SMD components) or by havin soldered into copper holes (through-hole components) on the circuit board.

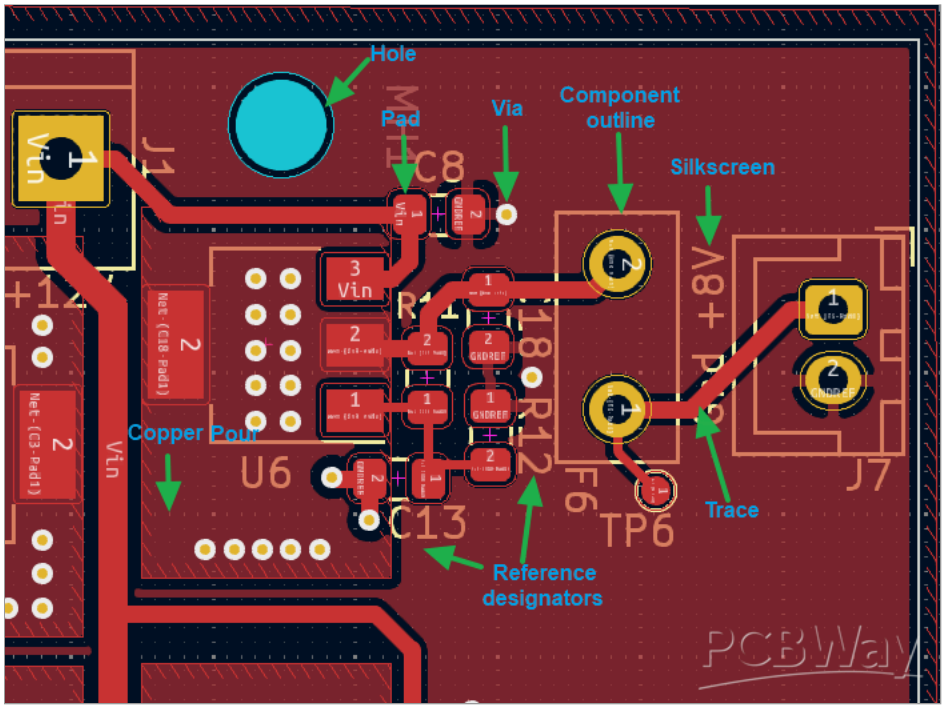
Pads and holes: Both are used to mount components to the circuit board and serve as locations to apply solder.

Silkscreen: Text and graphics printed on the surface of a PCB, including component markings, company logos, pa numbers, and other information used for manufacturing, assembly, and operational guidance.

References and designators: These tell the designers and assemblers the specific locations of components on th board. Each component has a unique reference designator, and these designators can be found in the design files software.

Soldermask: The outermost layer of a PCB, typically green, which protects the circuit board and prevents short ci during the soldering process.

Even though PCB layouts can be quite complex, understanding and identifying these features is crucial for designr manufacturing efficient and reliable circuit boards.



CAD drawing of a PCB layout

PCB Design Terminologys

Different ECAD applications may use slightly different terminology, but there are some terms that are universally across all PCB design applications:

Schematics: Schematics are like design blueprints that illustrate the electrical connections between components as connections for power and ground.

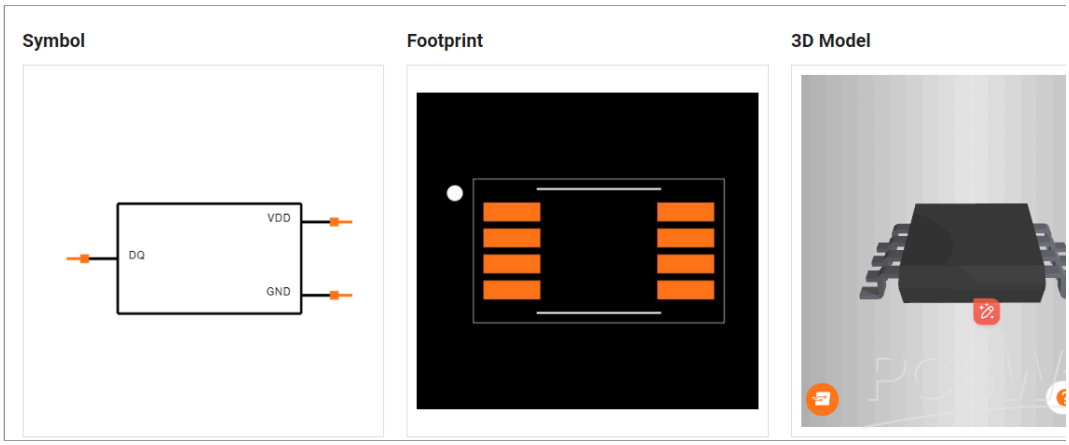
PCB Layout: This is the physical representation of a PCB, akin to a CAD drawing displayed above.

Nets: The PCB Net refers to the electrical connection path on the circuit board, connecting different components terminals in the circuit. These grids define the starting and ending points of electrical connections in the schemat diagram, which can be in various forms, such as bus connections or point-to-point connections. During the PCB la process, the grid will be actually physically connected through wires (called traces) to ensure the correct function performance of the circuit.

Schematic Symbol/CAE Symbol: This is typically a box with pins representing electrical connections available on component. Each pin represents a physical electrical connection on the actual component.

PCB Footprint/CAD Symbol: PCB footprint, or simply footprint, shows the arrangement of pads or holes on a com which is used for making electrical connections through routing on the PCB layout.

STEP File: A 3D model format used in mechanical modeling programs and advanced PCB design applications.



Example CAD data for DS18B20U+

Libraries: PCB libraries contain component models, each model comprising symbols, footprints, simulation mode STEP models.

SPICE: A general-purpose circuit simulation software package used in integrated circuit design and PCB design. I need to have simulation models of their components to run SPICE.

Annular Ring: Refers to the annular copper foil area between the edge of the through hole or via and the edge of t

Clearance: The minimum distance to prevent short circuits between signals, which is an important setting param PCB wiring.

Fanout: In the PCB layout process, Fanout refers to fan-out drilling, which means drilling short lines from the pads divided into two types: automatic and manual.

Aspect Ratio: Refers to the ratio between the thickness of a PCB and its smallest via diameter. It is best to keep t ratio low to improve plating quality and minimize the possibility of via failure.

DRC: Short for Design Rule Check, this is a software verification of the PCB layout, often used before production t that the PCB design does not contain any potential sources of error, such as small drill holes or traces that are to together.

Gold Fingers: Refers to the gold-plated connection contacts on the edge of the PCB, which are used to insert into connector or slot to achieve electrical connection. They are usually electroplated with gold to improve wear resist conductivity.

PCB Manufacturing Terminologys

Finally, here is a list of some basic PCB manufacturing terminologys designers may encounter.

PCB fabrication: Refers to the process of building a bare board from layered materials. This includes all parts of t board construction process.

PCB assembly: Refers to the process of populating devices to the bare board.

FR-4: This is not a specific material, but a class of materials used in circuit board design and manufacturing. This abbreviation indicates that the laminate is fire-resistant and meets NEMA Class 4 standards.

Gerber files, ODB++ files: These files are generated from the PCB layout and are used to generate the photolithog stencils, and are then used during the manufacturing process to etch the bare copper-clad PCB.

Netlist: Defines all electrical connections on a PCB in a human-readable format. This file is used to verify electric connections during manufacturability checks and to define test criteria for automated test equipment.

Pick-and-place: An automated assembly device used to grab and place components onto bare boards prior to sc

Soldering: Refers to three possible processes used to fix components to PCBs: manual soldering, selective solde wave soldering.

Tented Via: Refers to the use of soldermask to cover small vias in PCB to prevent solder filling or contamination. protects the vias and avoids short circuits or other problems during the soldering process.

Blind via: Located on the top or bottom surface of a printed circuit board. It has a certain depth but does not pene entire PCB. It is used to connect the surface layer circuits and the inner layer circuits below.

Buried via: Refers to the connection hole located in the inner layer of the printed circuit board, which does not ext the surface of the circuit board.

PTH: It is the abbreviation of Plated through hole. This kind of hole needs to be electroplated on the inner wall of through hole to make it conductive. It is usually used as a contact point for through-hole components and can als used as a via.

NPTH: It is the abbreviation of Non-Plated through hole, which refers to a hole without copper plating on the hole means that electrical connection cannot be made through the hole wall of this hole.

Last updated on 08/07/2024

Related articles

- Refuse conductive adhesive technology in pcb manufacture
- What a PCB Is Made Of
- What is your factory's standard of panel by tab-route?
- What is your requirements of V-cut?
- How to make my silkscreen clear and beautiful on PCB ?
- What diameter of via is better to have soldermask covering ?
- What is not correct for bottom text ?
- How to design the half (castellated) holes for PCB?
- PCB Manufacturing Terms Glossary
- Electronic Terms Glossary
- How to create a Centroid File(pick place) from Eagle, Altium,Sprint Layout and ORCAD?
- Generate Position File Centroid File(pick place) in Kicad
- How to generate BOM and Pick and Place File in Proteus 8.13
- How to Create a SMD footprint using the Allegro PCB Editor?
- How to Create a SMD pad using the Allegro Padstack Editor?
- How to Generate a Pick and Place File in .csv Format from Fusion 360?
- How to add a Board Outline to my Gerber Plots in DesignSpark PCB V11.0.0?
- PCB Panelization, Breakaway Rails, Fiducial Marks, Tooling Holes
- PCBWay Custom Design Rules

Comments (**0**)

This message here is only a suggestion supplement to the above content, and does not for trading operations su orders. If you have any urgent questions or order issues, please contact your sales rep.

Log in to post comments.

Upload photo

You can only upload 5 files in total. Each file cannot exceed 2MB. Supports JPG, JPEG, GIF, PNG, BMP

Post Comment

Capabilities

PCB Capabilities
PCB Assembly Service
Quick turn PCB Fabrication
SMD Stencils
Prototype PCBs
Quality Control

Products/Services

PCB Instant Quote
PCB FAQs
PCB Prototyping
Our Equipment
Blog/Engineering
PCB Products

Customer Support

Contact Us
Help Center
Sponsorships
Shared Projects
PCB Glossary
Referral Program

About us

About Us
Why Buy From Us?
Shipping Method Guide
Privacy Policy & Agreement
Refund Policy
Terms of Service

Payment Methods

Delivery Services

Verified by

Link Us on

PCB Prototype the Easy Way!

© 2024 PCBWay. All rights reserved.